LLOYD INSTITUTE OF MANAGEMENT AND TECHNOLOGY

(Affiliated to Dr. A.P.J. Abdul Kalam Technical University & Approved by AICTE, New Delhi)

Plot No. 3, Knowledge Park II, Greater Noida, Uttar Pradesh – 201306

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING & ALLIED BRANCHES

MINI PROJECT (BCC351 / BCS554 / BCS752)

Weekly Progress Log

Name: Rohan Pandey **Roll No.:** 2501720100117 **Section:** 2D

Project Title: *Real-Time Network Intrusion Detection System (NIDS) using Packet Inspection and Web-Based Alerting*

Project Guide: Mr. Vikas, Assistant Professor

Date	Work Done / Issues Faced	Mentor Remark & Sign
08–13 Sep 2026	Studied the mini project guidelines and the cyber-security domain. Selected the topic: a real-time network intrusion detection system. Learned what an IDS is and how it differs from a firewall.	
15–20 Sep 2026	Carried out the literature survey of Wireshark, Snort, Zeek and Scapy, and of the port-scan, SYN-flood and ARP-spoofing attacks. Prepared and submitted the project synopsis.	
22–27 Sep 2026	Synopsis approved and title locked. Designed the system architecture and drew the level-0 and level-1 data-flow diagrams and the use-case diagram. Fixed the detection thresholds.	
29 Sep–04 Oct 2026	Set up the development environment: installed Npcap, Python with Scapy, Node.js, Express and Socket.IO. Verified live packet capture. Issue: Windows capture needs Npcap and the correct interface.	
06–10 Oct 2026	Implemented packet capture with a filter and per-second TCP/UDP/ARP counts, and the port-scan rule with a sliding window and cool-down. First evaluation: live counts and a port-scan alert on the dashboard.	
13–18 Oct 2026	Built the Node.js and Express server to receive, validate and log alerts, and the live dashboard with a chart, alert table and colour-coded banner over Socket.IO. Added alert history that survives a page reload.	
20–25 Oct 2026	Implemented the SYN-flood rule (half-open handshake counting) and the ARP-spoofing rule (gateway trust table). Wrote ten automated unit tests covering all three rules with positive and negative cases; all pass.	
27–28 Oct 2026	Integration testing with self-generated attacks on an isolated host; measured detection latency and false alarms. Completed the sequence, component and deployment diagrams.	
29–30 Oct 2026	Wrote and formatted the full project report, ran the plagiarism check, and prepared the presentation. Pushed the source code to GitHub. Final evaluation and viva-voce.	

Student's Signature: _____ Guide's Signature: _____